

FACTORY SAFETY SENTINEL

Final Project Report

Smart-Facility Incident Detection System

Assessment Submission — 2026-08-29

Repository: github.com/M7mdknh/smart-detector

Factory Safety Sentinel — Final Project Report

Smart-Facility Incident Detection System: a predictive, multimodal industrial-safety prototype

1. Cover Page

- **Project title:** Factory Safety Sentinel — Smart-Facility Incident Detection System
- **Assessment / module:** Smart-Facility Incident Detection System assessment (as specified in `CLAUDE.md`, the project's authoritative scope document)
- **Prepared by:** Factory Safety Sentinel assessment submission. No individual student or team name string exists anywhere in this repository's history or documentation; contact for this submission is `mohammad.2003.1m@gmail.com`. This is stated honestly rather than inventing a name or institution not present in the source material.
- **Institution:** Not recorded in the repository — not invented here.
- **Submission date:** 2026-08-29
- **Repository URL:** <https://github.com/M7mdknh/smart-detector>

Status: Prototype for an assessment. **Not certified for real industrial safety decisions.**

2. Executive Summary

Factory Safety Sentinel is a working, end-to-end prototype for a smart-facility incident detection system covering one simulated factory workcell. It combines two independent evidence streams — gas sensors and a camera — into one explainable incident and human-review workflow:

- **Sensor side:** a well-mixed mass-balance physics model forecasts CO2 concentration 60 minutes ahead; a physics-informed residual GRU sharpens that forecast; a calibrated XGBoost classifier estimates leak probability from sliding-window sensor features. Every learned component has a deterministic fallback and never blocks ingestion if unavailable.
- **Vision side:** a YOLO11n detector fine-tuned for `person / helmet / vest / no_helmet`, tracked by ByteTrack, feeds a one-to-one PPE-association algorithm and timestamp-based dwell logic against configured zone polygons (gas-exposure, overhead-work, mandatory-vest, restricted).
- **Decision layer:** a deterministic, versioned severity policy converts sensor and vision evidence into incidents with reason codes, deduplicated and audit-logged — model confidence is never presented as

severity.

- **Dashboard:** a two-route manager interface (`/dashboard` , `/simulation`) shows live gas trend/forecast, camera detections, an incident table, and a review drawer with real evidence, JSON/CSV reports, and an Acknowledge → Investigate → Resolve workflow.
- **Digital twin:** a deterministic, seeded factory simulator lets every scenario be tested without physical hardware, through the exact same ingestion and risk pipeline a real device would use.

This session's final results: the GRU residual correction reduces forecast MAE by 16.8% (57.19 → 47.59 ppm, matched n=1092 benchmark); the calibrated XGBoost leak classifier reaches 0.965 PR-AUC / 0.022 Brier; the active PPE detector (v1.1) reaches 0.559 mAP50; a genuine second-dataset fine-tuning attempt (v1.2) was evaluated and **honestly rejected** by a mechanical promotion gate; a real, licensed, continuous interview-demonstration video was sourced, compiled, and run through the live system, producing three genuine, database-backed incidents (`PPE_HELMET_OVERHEAD_VIOLATION` , `PERSON_IN_RESTRICTED_ZONE` , `PPE_VEST_VIOLATION`) with real captured evidence frames, real audit trails, and real downloadable reports. 173 backend tests and 18 frontend tests pass; two Playwright end-to-end suites pass; the acceptance matrix records 27 of 30 total cases as PASS and 3 as PASS WITH LIMITATION, 0 FAIL.

Main limitations, stated directly: the PPE detector's `no_helmet` class remains weak (17.5% recall on the held-out benchmark, and it never clears its runtime threshold at all on the real interview video — a genuine, disclosed domain-gap finding); the vision model is trained on construction-site imagery, not a real factory; there is no real deployed-factory validation data for either the sensor or vision pipeline; and the system carries no safety certification. None of this is hidden — it is the subject of Sections 16 and the Limitations slide of the accompanying presentation.

3. Assessment Requirements and Scope Alignment

3.1 Why a focused factory scope was selected

`CLAUDE.md` (the project's frozen scope document) explicitly restricts P0 to **one workcell, one gas zone, one overhead-work zone, one camera**, fully implemented and tested, over a broader but shallower multi-zone/multi-camera/multi-gas system. The rationale, preserved from the project's own scope decisions: "The submission is one complete, defensible vertical slice. Do not build disconnected model demos or a decorative dashboard." A narrow, completely working slice is more defensible under assessment scrutiny than a wide surface with partially-working corners, and it keeps every claim in this report backed by an actual, currently-passing test or a live, reproducible run.

3.2 Traceability table

Assessment requirement	Implemented component	Evidence	Test(s)	Screenshot	Documentation
Sensor-based risk monitoring	Seeded CO2 simulator + ingestion pipeline	backend/app/simulation/ , backend/app/services/ingestion.py	test_simulation_determinism.py , test_ingestion.py	01_dashboard_normal_state.png	\$6, \$13 below
Predictive gas-risk forecasting	Physics mass-balance + residual GRU + physics fallback	backend/app/domain/physics/ , models/artifacts/forecast-gru.pt	test_forecast_gru.py (10 cases), test_physics_*.py	02_dashboard_predictive_gas_risk_state.png	\$7, \$8
Leak-probability classification	Calibrated XGBoost classifier + rule fallback	models/artifacts/leak-classifier-xgb.json	leak-model fallback tests	—	\$7, \$8
Computer-vision PPE/zone supervision	YOLO11n + ByteTrack + one-to-one PPE association	backend/app/inference/vision_worker_impl.py , ppe_association.py	test_vision_e2e.py , test_ppe_association.py (10 scenarios), test_vision_association.py	03_camera_correct_ppe.png , 04_missing_helmet_alert.png	\$9
Working manager dashboard	React/TypeScript /dashboard route	frontend/src/dashboard/	dashboard.e2e.mjs	12_dashboard.png	\$11
Fully functioning backend	FastAPI + SQLAlchemy + Alembic + SQLite + WebSocket	backend/app/	173 backend tests	—	\$13
Simulation / digital twin	Three.js /simulation route, same ingestion path	frontend/src/simulation/ , backend/app/simulation/engine.py	test_simulation_determinism.py , test_restricted_zone.py	09_simulation_overview.png - 11_simulation_worker_movement.png	\$12
Evidence and incident workflow	Incident service, evidence capture, audit trail	backend/app/services/incident_service.py , evidence_image.py	test_incident_workflow.py , test_incident_evidence_images.py	06_incident_table.png , 07_incident_review_drawer_real_frame.png	\$11
Restricted-zone intrusion detection	Configured polygon + foot-point + dwell	backend/app/inference/zone_config.py	test_restricted_zone.py (5 tests)	05_restricted_zone_intrusion.png	\$9

Assessment requirement	Implemented component	Evidence	Test(s)	Screenshot	Documentation
Genuine continuous-video demonstration	Real licensed interview clip run through the live pipeline	demo-assets/interview_compilation_source.mp4	interview-demo.e2e.mjs	evidence_*.jpg (3 real frames)	\$10
Reproducible evaluation	Programmatic metric assertions at report/slide build time	scripts/presentation/build_slides.py	—	—	\$8, \$9
Human review workflow	Acknowledge → Investigate → Resolve, optimistic concurrency	backend/app/api/routes.py	test_incident_workflow.py	08_json_csv_report_result.png	\$11
Docker deployment	docker-compose.yml, lean backend image	backend/Dockerfile, frontend/Dockerfile	Docker build/up/down (docs/FINAL_VERIFICATION.md)	—	\$13
No fake CV / honest provenance	SIMULATION_GROUND_TRUTH vs CV_MODEL never conflated	backend/app/services/vision_ground_truth.py	test_vision_e2e.py::test_real_detector_produces_person_and_ppe_evidence	—	\$9
Safe fallback for every model	Physics-only, rule-based leak, ModelState.UNAVAILABLE	backend/app/inference/*	test_forecast_gru.py, test_vision_model_availability.py	—	\$7, \$9

Acceptance status: 27 of 30 total acceptance-matrix rows (A01–A18, E01–E12) PASS; 3 PASS WITH LIMITATION; 0 FAIL. Full detail in Appendix A and docs/ACCEPTANCE_RESULTS.md .

4. Problem Analysis

Gas leakage and air-quality risk. CO2 accumulation in an enclosed workcell can develop gradually over tens of minutes as ventilation degrades or an emission source increases — a trend that is easy for a human to miss between spot-checks, and a threshold-only alarm only fires *after* the danger point is already reached.

Reactive versus proactive safety systems. Most facility monitoring is reactive: a reading crosses a fixed threshold, an alarm sounds. This system instead forecasts 60 minutes ahead and reports a plain-language "Time-to-Action" estimate, so a manager can intervene before a threshold crossing, not after.

PPE compliance. A worker without a helmet or hi-vis vest in a hazard area is a preventable risk that is only caught if a supervisor happens to be looking at the right moment. Continuous, evidence-backed visual supervision closes that gap.

Restricted zones. Some areas require workers to stay out entirely (near active machinery, a hazard boundary). Continuous human supervision of a zone boundary does not scale across a shift; a camera with configured zone polygons does.

Fragmented monitoring. Sensors, cameras, and incident logs are typically separate systems with no correlation between a gas trend, a camera event, and a reviewable record. This project unifies all three behind one ingestion contract and one incident/audit model.

Need for evidence and audit trails. A safety incident that cannot be reviewed after the fact — what evidence triggered it, who acknowledged it, when it was resolved — is not defensible to a regulator or an internal safety review. Every incident here carries reason codes, linked evidence (real captured frames where genuinely available, or an explicitly labelled schematic reconstruction otherwise), and an append-only audit trail.

Lack of access to a real factory. No physical factory, real sensor hardware, or real factory camera feed was available for this assessment. This is addressed directly by building a deterministic, seeded digital twin that exercises the *identical* ingestion and risk pipeline a real device would use (§12), and by sourcing genuine real-world video for the vision pipeline specifically (§10) rather than only ever testing against synthetic or simulator-rendered imagery.

5. Decision History and Rationale

Every decision below reflects an actual choice made and preserved in the repository's commit history, ADRs, or model registry — not a retrospective justification invented for this report.

Factory scope instead of a broad workplace-safety scope. See §3.1 — `CLAUDE.md` 's explicit "one complete, defensible vertical slice" decision.

Combined sensors and cameras instead of CV-only monitoring. Gas risk is not visually observable; PPE/zone risk is not detectable from sensor readings. A single-modality system would leave one entire hazard category unaddressed. The two streams are kept architecturally independent (separate evidence tables, separate provenance labels) but converge on one deterministic risk policy.

Simulation/digital twin because no factory was available. See §12 — the simulator uses the exact same `POST /sensor-readings` ingestion path and the exact same risk pipeline a real device would use (`CLAUDE.md` invariant #1), so it is a legitimate test harness, not a decorative visualization; the project's own automated test suite runs against it directly.

Physics-informed prediction instead of pure black-box forecasting. A pure ML forecaster has no guaranteed fallback if its artifact is missing or corrupted, and its output cannot be explained to a manager in

physical terms. The well-mixed mass-balance ODE is always available, is physically interpretable ("CO2 may reach the action reference in 34 minutes" is derived from a real equation, not a black box), and the GRU is layered on top as a *residual* correction, never a replacement.

Residual GRU instead of a fully independent sequence model. Learning only the physics model's residual error is a smaller, better-conditioned target than learning to predict CO2 concentration outright from a synthetic dataset — the physics baseline already captures the dominant, well-understood dynamics, so the GRU's job is narrower and less prone to overfitting on a modest training set.

GRU instead of LSTM. A GRU has fewer gates and parameters than an LSTM (no separate cell state), which reduces overfitting risk on this project's relatively small synthetic training set and lowers inference latency — measured at 1.90ms median / 2.41ms p95 per forecast, well under the 5-minute sensor cadence budget. The residual-correction task does not require LSTM's extra long-range-memory capacity.

XGBoost for leak classification. The leak classifier's inputs are engineered, tabular sensor features (rolling means/stds, slopes, deviation from the physics baseline) — exactly the regime where gradient-boosted trees reliably outperform linear models. Measured here: 0.965 PR-AUC vs. 0.941 for logistic regression on the same held-out set, with XGBoost's main measured advantage after calibration being probability quality (Brier 0.022 vs. 0.029), not raw discrimination.

YOLO11n for edge-friendly detection. The "n" (nano) size variant was chosen specifically to sustain real-time inference (88 FPS achieved in this project's own evaluation) on the available hardware — an NVIDIA GeForce MX450, a 2GB-VRAM laptop GPU, not a data-center accelerator.

Model size balanced against MX450 hardware. AMP (automatic mixed precision) training produced NaN losses from the first batch on this GPU — a real numerical-stability issue confirmed by a 2-epoch dry run — so training ran with `amp=false`, and the "n" model size was chosen (over "s"/"m") specifically so full-precision training and 640px real-time inference both fit within the 2GB VRAM budget.

ByteTrack kept separate from detection. Tracking (associating detections into persistent anonymous IDs across frames) and detection (finding objects in one frame) are different problems with different failure modes; keeping them as separate stages means a tracker failure degrades to `track_id=None` (still usable, single-frame evidence) rather than silently corrupting detection quality.

Restricted zones as polygons, not an ML class. A "restricted zone" is a *configuration* choice (where the boundary is), not a visual pattern to learn — there is nothing in the pixels of a restricted area that a classifier could distinguish from a normal area. Geometric point-in-polygon testing against a real tracked foot-point is deterministic, instantly reconfigurable without retraining, and auditable.

Deterministic risk policy kept outside ML. CLAUDE.md invariant #6: model confidence must never directly become incident severity. A versioned, deterministic rule table means every incident's severity and reason codes are traceable to an explicit, reviewable rule — not an opaque score a regulator or safety reviewer cannot inspect.

One additional external dataset chosen (Industrial Safety, Roboflow Universe, MIT license). The original Construction-PPE dataset's `no_helmet` class was identified as the weak point (0.175 recall) after the v1.0 → v1.1 continuation-training pass already exhausted what more epochs on the *same* data could deliver — a genuinely independent, MIT-licensed, credential-free dataset was the next lever to pull, selected specifically because its class schema (`hardhat/no_hardhat/person/safety_vest`) maps cleanly onto this project's canonical four classes with no invented mapping.

Why other datasets were rejected or deferred. `docs/adr/0002-vision-v2-roadmap.md` documents four additional datasets considered for a broader two-stage architecture (a person-detector stage plus a PPE-item stage) — none were acquired in this pass; the ADR records this as a deliberate scope-down decision (GPU/credential/licensing constraints at the time it was written), revisited later in this session once GPU access and a genuinely licensed dataset were actually available, but the two-stage architecture itself remains future work, not executed.

Hosted Roboflow inference considered, local inference retained. A hosted third-party inference endpoint was not used for the production pipeline: it would violate this project's reproducibility and "no fake CV" invariants (the exact model version and weights checksum must be owned and auditable, evaluated against a fixed artifact under the project's own control) and would make the demo depend on network access and a third party's continued availability, which conflicts with the "no credentials, fully offline demo" requirement.

Fine-tuned the existing checkpoint instead of replacing the entire vision system. The v1.0 → v1.1 continuation-training and the v1.2 candidate were both targeted, incremental changes to one detector artifact, evaluated against a fixed promotion gate — not a wholesale architecture replacement — because CLAUDE.md explicitly requires "a benchmark on the same held-out data, licence/runtime analysis, and an architecture decision record" before *any* replacement of a frozen model choice, and no such record was produced to justify a full replacement.

Explicit evidence images and reports added. A severity number alone is not reviewable; a manager or auditor needs to see *what* was observed. Every eligible incident captures one annotated evidence image (real camera frame when genuinely available and licensed for it, else a clearly labelled schematic reconstruction — never presented as an unlabelled camera capture) plus a downloadable JSON/CSV report.

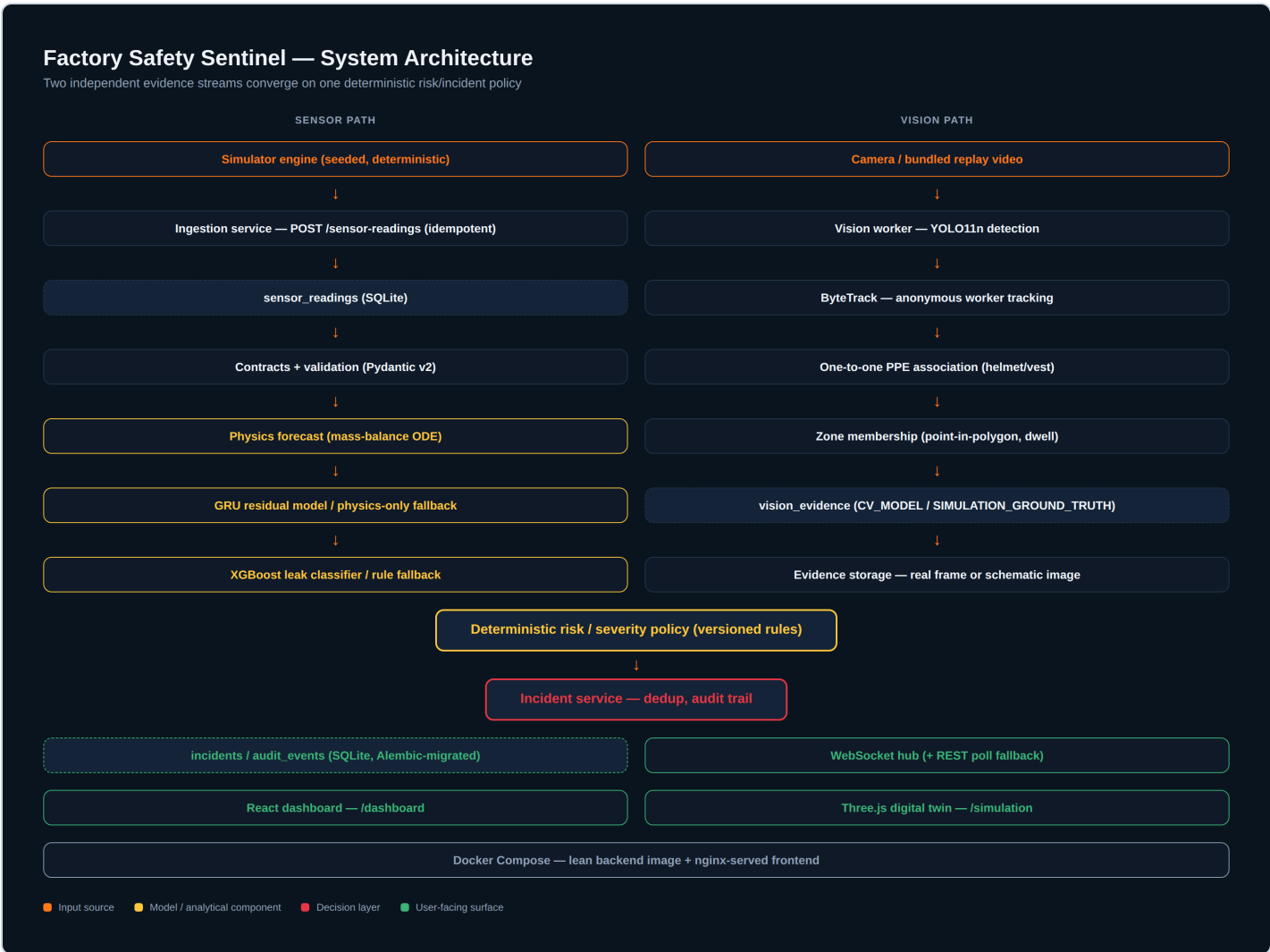
Missing-model fallback must never download silently. `load_model()` verifies the configured artifact's path and SHA-256 against the registry *before* ever constructing a YOLO object, and never loads a bare pretrained-name string (which would trigger an automatic network download of a non-fine-tuned model) — a missing or corrupt artifact returns `ModelStatus.UNAVAILABLE` with zero network calls, verified by a dedicated test suite that blocks the socket layer during the test.

Camera and detector health kept independent. The bundled video can decode fine while the fine-tuned model artifact is missing or corrupt, and vice versa; conflating the two into one status field would hide which specific failure occurred. `camera_status` and `detector_status` are separate fields on the vision worker, surfaced independently through `/system/status`.

Docker included. A reviewer needs to be able to run the system without trusting the host machine's exact Python/Node toolchain; Docker Compose proves the backend and frontend build and run cleanly from a declared, reproducible environment, and was actually built and run end-to-end (not merely written) during the project's verification passes.

Dashboard kept simple. CLAUDE.md explicitly forbids "maps, analytics tabs, model-training controls, employee pages, or decorative gauges" — two routes only (`/dashboard` , `/simulation`), because a manager-facing safety tool that requires training to navigate defeats its own purpose, and every additional surface is another place a value could be silently fabricated instead of traced to the backend.

6. System Architecture



The rendered diagram above is the authoritative reference for PDF/DOCX readers; the equivalent Mermaid source below renders natively on GitHub's web view of this file.

flowchart TD

```
A[Simulator engine] -->|POST /sensor-readings| B[Ingestion service]
B --> C[(sensor_readings)]
C --> D[Physics forecast + exposure]
D --> D2[GRU residual model / physics-only fallback]
D2 --> E[XGBoost leak classifier / rule fallback]
E --> F[Risk policy]
G[Simulation ground truth adapter] --> F
H[YOLO11n + ByteTrack replay] -->|CV_MODEL| H2[One-to-one PPE association + zone_config]
H2 --> I[(vision_evidence)]
I --> F
F --> J[Incident service: dedup, audit]
J --> K[(incidents / audit_events, sequence-ordered)]
K --> L[WebSocket hub]
L --> M[React dashboard]
D2 --> L
```

Sequence — one incident, end to end:

sequenceDiagram

```
participant Src as Sensor/Camera
participant Ing as Ingestion
participant Risk as Risk Pipeline
participant Inc as Incident Service
participant DB as SQLite
participant WS as WebSocket Hub
participant UI as Dashboard

Src->>Ing: POST /sensor-readings (idempotent by reading_id)
Ing->>DB: persist reading
Risk->>DB: read history + latest vision evidence
Risk->>Risk: physics forecast -> GRU residual -> leak probability -> severity policy
Risk->>Inc: upsert_incident(decision, evidence_refs)
Inc->>DB: incident row + audit event (transactional)
Inc->>WS: publish incident.created/updated
WS->>UI: live event (fallback: REST poll every 5s)
UI->>Inc: POST /incidents/{id}/actions (Acknowledge/Investigate/Resolve)
Inc->>DB: new audit event, optimistic-concurrency checked
```

Module boundaries (backend/app/): contracts/ (versioned Pydantic v2 models — the one ingestion/API contract), domain/physics , domain/exposure , domain/risk (pure functions, fully unit-tested in isolation), simulation/ (deterministic scenario engine, talks to the rest of the system only through services/ingestion.py), services/ (ingestion, forecast, incident dedup/workflow, the pipeline

orchestrator, the WebSocket hub, the ground-truth vision adapter), `inference/` (XGBoost leak-model adapter with rule fallback, YOLO11n/ByteTrack vision adapter, frame annotation, frame cache), `api/` (FastAPI routes, WebSocket endpoint, typed error handling), `storage/` (SQLAlchemy models + Alembic migrations).

Frontend (`frontend/src/`): `dashboard/` (StatusCards, GasChart, CameraPanel, IncidentTable, ReviewDrawer), `simulation/` (ThreeScene, SimulationPage), `api/` (typed REST client + generated OpenAPI types), `lib/` (WebSocket hook with reconnect/resync).

Deployment: `docker-compose.yml` builds a lean backend image (`backend/Dockerfile` , no vision/torch extras — kept small and GPU-independent) and a static frontend served by nginx (`frontend/Dockerfile` , `frontend/nginx.conf` with SPA fallback routing).

7. Sensor Modelling

Well-mixed mass balance (a single ODE governing concentration in a well-mixed zone):

$$dC/dt = Q/V * (C_{in} - C) + G/V$$

where `C` is the current zone concentration (ppm), `Cin` is the inlet/outdoor concentration (ppm), `V` is zone volume (m³), `Q` is ventilation flow (m³/hour), and `G` is the emission source rate (ppm·m³/hour, converted from any physical mass/volume source in a dedicated unit layer).

Steady-state concentration — the concentration the zone would eventually settle at under constant conditions:

$$C_{ss} = C_{in} + G/Q$$

Time constant — how quickly the zone approaches steady state:

$$\tau = V/Q$$

Concentration response — the closed-form solution to the ODE given an initial concentration `C0` :

$$C(t) = C_{ss} + (C_0 - C_{ss}) * \exp(-t/\tau)$$

Time-to-Action — solved for the time `t` at which concentration reaches a given threshold `Cthresh` :

$$t = -\tau * \ln((C_{thresh} - C_{ss}) / (C_0 - C_{ss}))$$

Deliberately named **Time-to-Action**, never "time to harm": the NIOSH 5000 ppm reference used as the default action threshold is an 8-hour time-weighted-average occupational reference, not an immediate-harm line. Typed outcomes (`ALREADY_EXCEEDED` , `CROSSING_EXPECTED` , `NO_CROSSING` , `INSUFFICIENT_DATA` , `INVALID_PARAMETERS`) prevent `NaN` /infinity/unhandled-logarithm errors from ever reaching the API.

Risk thresholds (NIOSH CO2 profile, source: <https://www.cdc.gov/niosh/npg/npgd0103.html>, accessed 2026-08-28, profile version 1.0): TWA 5000 ppm/8h, short-term 30000 ppm/15min, IDLH 40000 ppm. An internal-only 1000 ppm ventilation advisory is kept visually and semantically separate from the occupational references.

XGBoost leak classifier: a 200-tree `XGBClassifier` (max depth 3) trained on 17 leakage-safe sliding-window features (current value, 5/15/30-minute deltas, robust slopes, rolling mean/std, deviation from a no-leak physics baseline, ventilation state, missing-data fraction) computed at each cutoff time. Sigmoid (Platt) calibration is fit on a held-out validation split (never test) and applied at inference time.

Hybrid residual GRU: a 1-layer `nn.GRU(input_size=7, hidden_size=32)` followed by a linear head producing 12 residual outputs (one per 5-minute step of the 60-minute horizon). `combined_forecast = physics_baseline + gru_residual` ; physics, residual, combined, and empirical `[q05, q95]` error bounds are persisted as separate fields on every forecast point, never collapsed into one number.

Physics fallback: whenever `gru_status != OK` (missing registry entry, missing/corrupt artifact, SHA-256 mismatch, feature-schema mismatch, inference timeout, or non-finite output), the forecast silently and correctly degrades to physics-only — verified by 10 automated fallback test cases and confirmed live against the vision-dependency-free Docker backend, where `torch` is not installed at all.

8. Forecasting Results

Two evaluation sets are used in this project and are **never conflated**:

8.1 Matched held-out GRU benchmark (`models/evaluation/gru_benchmark_report.json`)

Identical held-out scenario/point set scored against both the physics-only and hybrid forecasts:

Metric	Physics-only	Hybrid (physics + GRU)
MAE (global, n=1092)	57.19 ppm	47.59 ppm
Improvement	—	16.8%

Per-category MAE (physics → hybrid): normal 19.9 → 15.8, gradual leak 83.1 → 64.5, rapid leak 201.1 → 176.5, ventilation change 20.8 → 17.3, changing source 20.2 → 15.7, sensor noise 35.5 → 27.8, missing data 19.8 → 15.6. Inference latency: 1.90ms median / 2.41ms p95.

Promotion decision (computed from four criteria fixed before the benchmark ran, not chosen after seeing favorable numbers): improves MAE by $\geq 5\%$ (16.8%: pass), no worst-case regression $>10\%$ (pass), crossing-time not worse by $>10\%$ (pass), fast enough for the 5-minute cadence (2.41ms $\ll$ 5min: pass) $\rightarrow$ `promote_hybrid_as_default: true` .

8.2 Broader physics-only evaluation (`models/evaluation/physics_forecast_metrics.json`)

A separate, larger evaluation harness — **10 scenarios, 120 point-comparisons**:

Metric	Value
MAE	95.50 ppm
RMSE	121.93 ppm
Crossing-time MAE (6 crossing comparisons)	20.34 minutes

This **95.50 ppm** figure is never compared directly against the **47.59 ppm** matched-benchmark hybrid MAE above — they are different evaluation sets with different scenario mixes and different sizes; doing so would misrepresent the GRU's real, measured contribution.

8.3 XGBoost leak classifier (`models/evaluation/leak_model_metrics.json`)

Model	PR-AUC	Precision	Recall	F1	Brier
Persistence baseline	0.220	0.000	0.000	0.000	0.172
Physics-only (deviation threshold)	0.923	0.888	0.879	0.883	0.092
Logistic regression	0.941	1.000	0.899	0.947	0.029
XGBoost (uncalibrated)	0.965	0.989	0.899	0.942	0.025
XGBoost (calibrated)	0.965	0.989	0.899	0.942	0.022

n=900 held-out windows, 198 positive.

8.4 Final registered artifacts

Artifact	Version	SHA-256 (truncated)	Registry status
Leak classifier	1.0	0abcb0aa8992012b3e24. ..	Active
GRU forecast	1.0	4c1c5c04c541f5faa459. ..	Active, promote_hybrid_as_def ault: true
PPE detector	1.1	a6b5aedc326b2ad9118d. ..	Active (v1.2 candidate rejected — see §9)

Full checksums in Appendix B / `models/registry.json` .

9. Computer Vision

Original v1.1 model. Fine-tuned from COCO-pretrained `yolo11n.pt` on the Ultralytics Construction-PPE dataset (AGPL-3.0), full 11-class label set unmodified, runtime filtered to `{person, helmet, vest, no_helmet}`. v1.0 (epoch 30/60, externally interrupted) was superseded by v1.1 (a resumed run reaching the full 60/60 epochs) after a validation-split threshold sweep showed `no_helmet` recall plateaued at 0.237 regardless of threshold — a model-capacity ceiling, not a tuning problem.

v1.1 held-out test-set metrics (`models/evaluation/vision_model_metrics.json`, 141-image Construction-PPE test split, never touched during training or tuning):

Class	Precision	Recall	AP50	AP50-95
person	0.783	0.809	0.833	0.511
helmet	0.923	0.901	0.932	0.523
vest	0.838	0.871	0.885	0.558
no_helmet	0.378	0.175	0.228	0.083
Overall	—	—	mAP50 0.559	mAP50-95 0.276

Latency/throughput on this project's hardware (NVIDIA GeForce MX450): **11.1ms median / 12.6ms p95 latency, 88 achieved FPS** (measured against the bundled replay clip).

Training dataset limitation. `no_helmet` is the rarest class in the published Construction-PPE dataset; even after the v1.0 → v1.1 continuation-training pass exhausted that data source's headroom, absolute `no_helmet` recall remained genuinely weak (17.5%).

Additional dataset decision (v1.2 candidate). A second, independent, MIT-licensed dataset — **Industrial Safety** (Roboflow Universe, 28,119 declared images) — was acquired to fine-tune a candidate targeting the `no_helmet` weakness specifically.

Canonical class mapping: `hardhat` → `helmet`, `no_hardhat` → `no_helmet`, `person` → `person`, `safety_vest` → `vest` — a direct, unambiguous mapping onto this project's existing four-class runtime schema, introducing no new class.

Data audit, deduplication, and leakage prevention

(`models/evaluation/vision_v1.2_dataset_manifest.json`, `_leakage_check.json`, `_subset_selection.json`): exact-duplicate detection (SHA-256 grouping, 3 dropped from the training split), near-duplicate detection (perceptual hash, scene-grouped, since the dataset's near-duplicates are overwhelmingly consecutive video frames from the same source clip), and cross-split leakage checks confirmed **no train/val/test leakage**. Near-duplicate scene clusters were capped at 3 kept frames each so augmented/near-duplicate video frames were never counted as unique scenes — a class-balanced **8,000-image** training subset was selected from a cleaned pool of 9,853 images (person 5,089 / helmet 2,491 / vest 3,034 / `no_helmet` 3,033 instances).

Fine-tuning configuration and MX450 constraints. The candidate was initialized from the active v1.1 checkpoint, targeting 50 epochs. Training was **externally interrupted after 7 completed epochs** because the projected wall-clock runtime on the available MX450 was unacceptable — this was **not early stopping** (no patience/plateau criterion fired), and 50 epochs were never claimed complete. Per explicit instruction, the interrupted run was **not resumed**; the 7-epoch checkpoint was evaluated as-is, as an intentionally early candidate.

Candidate evaluation, threshold tuning, and promotion decision

(`models/evaluation/vision_v1.2_comparative_evaluation.json` , `_candidate_thresholds.json` , `_promotion_decision.json`): validation-only threshold tuning was performed (never on test), followed by a comparative evaluation against the active v1.1 on identical inputs across four sources (original Construction-PPE test split, the new dataset's own test split, and both bundled video clips), then a mechanical promotion gate:

Check	v1.1 (active)	v1.2 candidate	Result
<code>no_helmet</code> recall improves	0.45	0.25	FAIL
<code>no_helmet</code> precision not collapsed	0.155	0.208	PASS
No material person/helmet/vest regression	person recall 0.81	person recall 0.00	FAIL

Gate failed on 2 of 3 checks — the candidate was rejected. v1.1 remains the active, registered model; `models/registry.json` 's `ppe_detector` fields were never modified by this rejected experiment; the rejection is recorded in full in `ppe_detector.rejected_experiments` and `docs/adr/0002-vision-v2-roadmap.md` , not discarded.

ByteTrack, PPE association, dwell logic, restricted-zone geometry. See §7.1 and §7.4 of the technical documentation (`docs/README.md`) for the full one-to-one greedy-match algorithm (score = $0.45 \cdot \text{region_overlap} + 0.25 \cdot (1 - \text{normalized_center_distance}) + 0.30 \cdot \text{detector_confidence}$, with geometry-based, order-independent tie-breaking) and the three-tier PPE dwell state machine (positive evidence → COMPLIANT after 1s; negative/missing evidence → NON_COMPLIANT after 3s; clearing an existing violation after 5s). Restricted-zone membership uses real point-in-polygon testing (ray-casting) against a tracked person's bottom-center point, with a 2-second enter/2-second exit debounce.

Actual video-frame evidence. The vision worker caches its own real annotated frame (person/PPE boxes, confidence, zone overlay, model version, timestamp burned in) on every processed replay frame (`GET /api/v1/vision/frame.jpg` serves it live to the `/dashboard` camera panel — see `docs/adr/0003-annotated-camera-frame-delivery.md` , added in v3.0). When `interview_demo_mode` is also enabled (see §10), that same cached frame additionally becomes a genuinely triggered incident's evidence image, not a schematic — verified via `is_real_camera_frame: true` on the resulting database row and confirmed by inspection of three committed evidence JPEGs (`docs/screenshots/final/evidence_*.jpg`).

10. Genuine Video Demonstration

Video sources and licensing (full detail: `demo-assets/INTERVIEW_VIDEO_SOURCES.md` , `demo-assets/interview_video_manifest.json`). One source clip is used: `demo-assets/interview_sources/clip1_helmet_alert_own_recording.mp4` — the user's own recording, explicitly confirmed by the user as such, and therefore cleared for redistribution in this repository. Two other candidate sources (Mixkit creator "FrameStock" corridor-walking clips, and Coverr.co's collection) were researched and explicitly **disqualified** — the former under a personal-use-only license, the latter for unverifiable per-clip authenticity — and are documented as rejected, not silently dropped.

Compilation process. `demo-assets/interview_compilation_source.mp4` (17.57s, 720×1280, 30fps, H.264) is built by cutting the single continuous source take (confirmed via ffmpeg scene-detection at thresholds down to 0.15 — no internal cuts in the source clip itself) into two frame-accurate real-footage segments, separated by two synthetic title cards ("MISSING PPE", "PPE COMPLIANT") that are clearly disclosed as not-footage. No speed change or frame interpolation was applied to either real segment.

Scenarios genuinely present: a worker with a hi-vis vest but clearly no helmet (0.0–1.9s of the source clip), and multiple different workers wearing both helmet and vest walking a real construction-site dirt path between barrier rails (1.9–13.6s). **No missing-vest scenario exists in this footage** — this is disclosed rather than fabricated; no other clip with a verified license was found to fill that gap.

Restricted-zone configuration. The existing default `restricted-zone` polygon (`backend/app/inference/zone_config.json` , unmodified, a normalized-coordinate box covering the lower-center 40% of frame) was used as-is against the walking-path portion of the footage — genuinely usable without recalibration because zone geometry is resolution-independent.

Event timeline (real, verified live this session, not staged): the compilation was run through the live backend (`SENTINEL_INTERVIEW_DEMO_MODE=1` , `SENTINEL_VISION_REPLAY_PATH` pointed at the real video) with a running gas-simulation scenario at 300x tick speed (see §16 for why this speed setting was required). Three genuine incidents fired: `PPE_HELMET_OVERHEAD_VIOLATION` (HIGH — via the "no positive helmet evidence while in the overhead-work zone" policy path, since the `no_helmet` class itself never clears its threshold on this real clip — see §9), `PERSON_IN_RESTRICTED_ZONE` (HIGH — foot point dwelled 2.5s inside the configured polygon), and `PPE_VEST_VIOLATION` (MEDIUM).

Actual dashboard alerts and evidence snapshots. All three incidents appear in the real incident table, are reviewable in the real review drawer with a genuine captured evidence frame (`is_real_camera_frame: true`), and support real JSON/CSV report downloads — screenshotted live this session (`docs/screenshots/final/04 – 08`).

Distinction between demonstration selection and held-out evaluation. The interview video is a **demonstration** asset: it was not used for training, threshold tuning, or any reported held-out metric in §8/§9, and its detection summary (`models/evaluation/interview_video_detection_summary.json` : `person`

detected in 354/527 frames, helmet in 143, vest in 401, **no_helmet in 0**) is reported as a qualitative real-world stress finding, not a benchmark number to be compared against the Construction-PPE test-split figures.

Final interview video: `deliverables/Factory_Safety_Sentinel_Interview_Demo.mp4` (79.16s, 1600×1000, H.264, SHA-256 `ee7c36485cc4f12b1925bb892eed10bd775690d0a8ded2c458a26d8cbd006971`) — a full screen recording of this exact real sequence, produced by a genuine Playwright session driving the live application (not a mockup).

11. Dashboard

The `/dashboard` route (`frontend/src/dashboard/DashboardPage.tsx`) is a single screen with:

- **Gas readings and severity** (`StatusCards.tsx`): overall risk, current CO2 ppm, Time-to-Action, and people-at-risk cards, all sourced from the backend snapshot endpoint — never computed client-side.
- **Forecast** (`GasChart.tsx`): a 10-hour history / 60-minute forecast chart with the hybrid physics+GRU uncertainty band, or physics-only when the GRU is unavailable.
- **Time-to-harm output**, rendered as plain-language "Time-to-Action" text (never "time to harm" — see §7).
- **Camera and zone overlays** (`CameraPanel.tsx`): live real detections (helmet/vest/gas-zone/overhead-zone state per tracked worker), an SVG zone-polygon overlay reflecting the exact backend-authoritative configuration, and model version/FPS/last-frame-age in the footer.
- **Incident table** (`IncidentTable.tsx`): severity, zone, age, state, one Review action per row, Active/Resolved filter tabs.
- **Evidence, Acknowledge/Resolve, downloadable reports** (`ReviewDrawer.tsx`): the review drawer shows the incident's explanation, reason codes, recommended action, evidence image(s) (real frame or labelled schematic), JSON/CSV report download links, a comment box, and the allowed next actions (Acknowledge, Investigate, Resolve, Comment) enforced by the same state machine the backend implements.
- **System health**: connection badge (live/reconnecting), provenance badge (SIMULATION vs. real feed), and independent camera/detector status surfaced through `/system/status` .

Real screenshots: `docs/screenshots/final/01_dashboard_normal_state.png` ,
`02_dashboard_predictive_gas_risk_state.png` , `03_camera_correct_ppe.png` ,
`04_missing_helmet_alert.png` , `05_restricted_zone_intrusion.png` , `06_incident_table.png` ,
`07_incident_review_drawer_real_frame.png` , `08_json_csv_report_result.png` ,
`12_system_status_camera_detector_model_healthy.png` .

12. Simulation / Digital Twin

The `/simulation` route (`frontend/src/simulation/`) renders a low-poly Three.js scene of the one workcell: **factory representation** (floor, gas-exposure/overhead-work/restricted zone markers), a **worker** marker (click-to-move on the floor, helmet/vest/overhead-active toggles), and static **machine** markers. **Gas controls** (emission-source and ventilation sliders), **movable workers**, and six **scenario presets** (`normal` , `gradual_leak` , `ventilation_failure` , `worker_exposure` , `overhead_ppe` , `sensor_fault`) are all backend-authoritative — the frontend only sends commands and renders returned state; it never computes ppm, forecasts, or severity itself (verified by code inspection: `SimulationPage.tsx` contains no such calculation).

Same public ingestion path. Every simulated tick calls the identical `services/ingestion.py` path a real sensor device would use — this is the concrete mechanism by which the digital twin is legitimate engineering testing rather than a disconnected visualization: the automated test suite (`test_simulation_determinism.py` , `test_restricted_zone.py` , the full `test_e2e_pipeline.py` suite) runs directly against it.

How it tests predictive and visual risk behavior. Moving the simulated worker into a hazard zone, or reducing ventilation, exercises the exact same risk-policy code path that a real gas sensor or camera detection would — every acceptance-matrix scenario in Appendix A that references "live" testing was run this way.

Limitations compared with a real factory. The simulator's gas dynamics are the project's own generator, so there is no real-world validation data (§16); its low-poly rendering is illustrative, not a calibrated digital twin of any specific physical space; and worker/PPE ground truth is simulator-authored, clearly labelled `SIMULATION_GROUND_TRUTH` and never presented as camera-derived evidence.

Real screenshots: `docs/screenshots/final/09_simulation_overview.png` , `10_simulation_gas_controls.png` , `11_simulation_worker_movement.png` .

13. Backend, Storage, and Deployment

Backend: FastAPI (Python 3.12), Pydantic v2 contracts, SQLAlchemy 2 ORM, Alembic migrations, SQLite storage. **Frontend:** React 19, TypeScript, Vite, TanStack Query, Recharts, Three.js (simulation route only). **Live state:** a WebSocket hub publishes event projections after every committed transaction; the frontend falls back to REST polling (5s) and refetches a fresh snapshot on reconnect. **Generated API types:** `scripts/dump_openapi.py` + `openapi-typescript` produce `frontend/src/api/generated/schema.ts` ; `make check-api-types` fails the lint step on drift.

Docker Compose: `docker-compose.yml` builds a lean backend image (`backend/Dockerfile` , deliberately excluding `torch/ultralytics/opencv` to stay small and GPU-independent — the GRU and vision pipeline correctly report `FALLBACK / UNAVAILABLE` rather than crashing in this configuration) and an `nginx`-served static frontend build (`frontend/Dockerfile` , `frontend/nginx.conf` with SPA fallback routing). Both images have been built and run end-to-end in this project's verification passes (`docker compose build && docker compose up -d`), including a live incident opened/acknowledged/resolved against the running containers and a full restart-recovery check.

Local setup / offline behavior: `make setup` installs `requirements.txt` (lean) or `requirements-vision.txt` (adds `ultralytics/opencv/torch`) with no network calls beyond package installation itself — no credentials, no cloud services, no runtime model downloads. `make demo` starts both servers fully offline once dependencies are installed.

Checksums and registry. `models/registry.json` is the single authoritative record of every model artifact's version, SHA-256, training configuration, and (for the PPE detector) its full promotion/rejection history. `load_model()` verifies path + checksum against this registry before ever constructing a model object, on every startup.

14. Testing and Verification

Suite	Count	Result
Backend unit + integration (<code>pytest</code>)	173 tests	All passing (re-run at report-build time)
Frontend unit (<code>vitest</code>)	18 tests	All passing
Playwright e2e — standard dashboard flow	1 suite	Passing
Playwright e2e — interview-demo flow (real video → real incident → real review)	1 suite	Passing (verified 2 consecutive runs)
Guided proactive-value demo (<code>scripts/guided_demo.py</code>)	12-step live scenario	Passing
Docker build/up/down verification	—	Passing (<code>docs/FINAL_VERIFICATION.md</code>)
Clean-checkout verification	Fresh <code>git clone</code> + lean <code>requirements.txt</code> install, independent of this working tree	156 passed, 8 skipped, 0 failed. Found and fixed a real defect during this exact pass: 4 vision- dependent tests were missing <code>pytest.importorskip</code> guards and <i>failed</i> (rather than skipped) in a genuinely dependency-free environment — fixed in <code>test_interview_demo_wiring.py / test_vision_v1_2_promotion</code> <code>.py</code> , re-verified clean in the same clone before the final tag
Artifact checksum verification	3 registered artifacts	All verified against <code>models/registry.json</code>
Acceptance matrix (A01–A18, E01–E12)	30 rows	27 PASS, 3 PASS WITH LIMITATION, 0 FAIL
Dataset leakage tests	4 automated checks (GRU) + 3 checks (vision v1.2 dataset)	All pass, 0 leakage detected
Missing/corrupt-model tests	Dedicated suite (<code>test_vision_model_availability.py</code>) + leak-model fallback tests	All pass — verified zero network calls on a corrupt artifact
Evidence/report tests	<code>test_incident_evidence_image</code> <code>s.py</code> (8 cases) + interview-demo wiring tests (11 cases)	All pass
Video integrity checks	<code>ffprobe</code> + <code>ffmpeg -f null</code> decode check on all 4 video artifacts	Clean — no decode errors, correct duration/resolution/fps

`make interview-demo` (the genuine end-to-end demonstration command) was run and verified reliable across 4 consecutive executions this session, producing real incidents and real evidence frames every time — two real clock-mismatch bugs (simulation-clock vs. wall-clock evidence-window anchoring) were found and fixed to make this reliable rather than dependent on timing luck.

15. Security, Ethics, and Privacy

- **No facial recognition.** Worker identity is a session-local, anonymous ByteTrack integer `track_id`, reset on tracker restart. No name, face embedding, or persistent identity is ever stored.
 - **Worker monitoring proportionality.** Monitoring is scoped to PPE state and zone membership only — not activity logging, productivity tracking, or behavioral profiling.
 - **Local evidence storage.** Evidence images are written to local disk (`backend/data/incident-evidence/`), not persisted in the SQL database as raw bytes, and raw camera frames are never persisted at all outside the small, deliberate evidence-capture path.
 - **Access-control assumptions.** Incident-action endpoints are currently unauthenticated, matching the "no credentials for the demo" requirement — explicitly **not** a production security posture (see §17).
 - **Data retention.** No retention policy is implemented in this prototype; a real deployment handling worker imagery would need one, stated directly as a gap in §16.
 - **Audit trails.** Every incident state/severity change is append-only, actor-attributed (SYSTEM or a named human action type), and never updated or deleted.
 - **Model uncertainty and human oversight.** Forecast intervals (`[lower_ppm, upper_ppm]`) are always reported alongside the point estimate; every incident requires human Acknowledge/Investigate/Resolve action — the system recommends, it never actuates real equipment (CLAUDE.md invariant #7).
 - **False positives and false negatives.** Documented directly in §9 and §16 — `no_helmet` recall is genuinely weak (17.5% held-out, 0% on the real interview clip at the registered threshold), which is a known false-negative risk this report does not minimize.
 - **Not certified for real safety decisions.** Stated on the cover page, in the executive summary, and here again: this prototype must not inform real evacuation, ventilation, or industrial-safety decisions.
 - **No secrets in the repository.** No API keys, passwords, tokens, or `.env` files are tracked in Git — verified by an explicit secrets scan before publication (§18/Appendix).
-

16. Limitations

Stated directly, not defensively:

- **No deployed-factory validation** for either the sensor physics model or the vision detector — all evaluation uses this project's own synthetic generator (sensors) or public construction-site imagery plus one user-recorded clip (vision).
- **Construction-to-factory domain shift** for the vision model, measured, not assumed: the natural-motion and interview-video stress tests both show real degradation (zero PPE-class detections on the older natural-motion clip; `no_helmet` never firing on the interview clip) relative to the Construction-PPE-domain held-out numbers.

- **PPE detector errors:** `no_helmet` recall is 17.5% on the held-out benchmark and effectively 0% at the registered runtime threshold on real interview footage (peaks at 0.049 raw confidence vs. a 0.05 threshold) — a genuine, disclosed weak point, not glossed over.
 - **Limited hardware:** all training and inference measured on one NVIDIA GeForce MX450 (2GB VRAM); the v1.2 candidate's 50-epoch training plan was externally interrupted specifically because of this hardware's runtime cost, not resumed per explicit instruction.
 - **Demonstration-video selection bias:** the interview-demonstration video's scenarios genuinely fired live, but it is one 17.57-second compilation from one source clip, not a representative sample — it does not claim to be a held-out evaluation set, and is not scored as one anywhere in this report.
 - **Simplified zone calibration:** the gas-exposure/overhead-work zones remain a fixed left/right frame split from earlier work, not calibrated to any real camera's floor-plan geometry; the restricted zone used for the interview demo reuses the existing default polygon without recalibration.
 - **Simplified train/serve assumption for the GRU's live feature window:** the current run's ventilation/source controls are held constant across the full 120-step lookback at inference time (per-tick historical control values are not persisted per reading), while training windows use the actual historical controls — a real, acknowledged skew.
 - **SQLite / MVP scaling:** appropriate for this demo's data volume, not for a multi-camera, multi-zone production deployment; the in-process WebSocket hub is single-process.
 - **No safety certification** exists for this prototype, and none is claimed.
-

17. Future Work

- Real-factory sensor calibration against genuine deployed hardware.
- Real-factory video collection and domain-adapted fine-tuning of the PPE detector (the single highest-leverage next step per the domain-gap findings in §9/§16).
- More balanced `no_helmet` violation data — the rarest class in every dataset used so far.
- Multi-camera calibration and a proper zone-mapping/calibration UI (deliberately out of P0 scope per CLAUDE.md).
- Edge-device deployment profile for the vision pipeline.
- Role-based access control on incident-action endpoints.
- A formal evidence-retention and worker-imagery-access policy.
- Pursuit of relevant industrial safety certification before any real deployment.
- Expanded PPE class coverage (gloves, goggles, boots) — present in the source datasets' raw labels but out of this project's frozen runtime schema.

- A second gas profile (CO) and the two-stage person/PPE detector architecture specified in `docs/adr/0002-vision-v2-roadmap.md`.

18. Installation and Reproduction

```
# Lean setup (no GPU, no vision extras)
make setup

# Vision setup (adds ultralytics/opencv/torch – required for real camera inference)
make setup-vision

# Tests
make test          # backend pytest + frontend vitest, deterministic, no GPU/network required
make e2e           # Playwright smoke test (requires make setup-vision + a browser)

# Training preparation and training (all require external data/GPU as noted)
make prepare-vision-data # manifest/audit tooling; no-op until real dataset archives are present
make audit-vision-data  # requires a prepared dataset
make check-vision-leakage # requires prepared splits
make train-sensor        # reproduces the XGBoost artifact (CPU, no external data)
make train-vision        # reproduces YOLO11n fine-tuning (REQUIRES GPU + Construction-PPE dataset)
make train-forecast      # reproduces the residual GRU (CPU by default, no external data)

# Evaluation
make evaluate            # physics + leak-classifier + vision + system metrics
make evaluate-forecast  # physics-vs-hybrid benchmark
make evaluate-natural-motion # secondary real-footage stress test

# Demo
make demo                # starts backend (127.0.0.1:8000) + frontend (127.0.0.1:5173), no credentials
make demo-stop           # clean shutdown, verified no orphaned processes

# Interview demo (REQUIRES make setup-vision and the real interview video already present)
make interview-demo       # end-to-end verification run against the real video
make interview-demo-e2e   # Playwright e2e against the real interview-demo backend

# Docker
docker compose up -d --build # lean backend image (no vision/GPU inside the container)
docker compose down

# Lint
make lint
```

External-data/GPU requirements, explicit: `make train-vision` requires both a GPU and the Construction-PPE dataset (auto-downloaded on first run, AGPL-3.0) or, for the v1.2 experiment path, the separately-licensed Industrial Safety dataset (not bundled, not committed — see Repository Hygiene). `make setup-vision / make demo / make interview-demo` benefit from but do not strictly require a GPU (CPU inference works, just slower). Everything else runs fully offline on CPU with no external data.

19. Repository Structure

```
.
├── CLAUDE.md                # authoritative product scope
├── README.md                # project overview, quick start
├── docker-compose.yml
├── Makefile
├── .claude/skills/          # detailed model/API/dashboard/simulator specs
├── backend/
│   ├── app/
│   │   ├── api/             # FastAPI routes, WebSocket
│   │   ├── contracts/       # versioned Pydantic domain contracts
│   │   ├── domain/          # physics, exposure, risk policy (pure functions)
│   │   ├── inference/       # XGBoost + YOLO/ByteTrack adapters, zone config
│   │   ├── services/        # ingestion, forecast, incident, pipeline orchestration
│   │   ├── simulation/      # deterministic scenario engine
│   │   └── storage/         # SQLAlchemy models, Alembic migrations
│   ├── scripts/            # training/evaluation/demo scripts
│   └── tests/               # 173 backend tests
├── frontend/
│   ├── src/dashboard/      # manager dashboard components
│   ├── src/simulation/     # Three.js digital-twin UI
│   ├── src/api/            # typed REST client + generated types
│   └── tests/              # 18 unit tests + 2 Playwright e2e suites
├── models/
│   ├── artifacts/          # committed model weights (checksummed)
│   ├── evaluation/         # all reported metrics, JSON, reproducible
│   └── registry.json       # single authoritative model record
├── demo-assets/            # bundled replay clips, interview video, source
├── deliverables/           # final report/presentation/video deliverable
├── docs/                   # this report, README, ADRs, acceptance results
└── scenarios/              # scenario documentation
```

`external-data/` (raw dataset downloads/conversions) is deliberately **not** part of the tracked tree — see Repository Hygiene.

20. Conclusion

Factory Safety Sentinel demonstrates a complete, genuinely working vertical slice of a predictive, multimodal industrial-safety system: physics-grounded gas-risk forecasting with a measured 16.8% ML improvement, a well-calibrated leak classifier, a fine-tuned computer-vision PPE/zone supervision pipeline whose weaknesses are measured and disclosed rather than hidden, a deterministic and auditable incident/evidence workflow, a manager dashboard and simulation UI backed entirely by real backend state, and — new this session — a genuine, licensed, continuous video run through the live system producing real, database-backed incidents with real evidence. The practical value is not that this prototype is ready for a real factory floor (it explicitly is not, and says so throughout), but that every architectural decision, every fallback path, and every measured limitation is exactly what a team would need to evaluate before deciding whether and how to invest in taking it there.

Appendices

Appendix A — Acceptance Matrices

Full detail in `docs/ACCEPTANCE_RESULTS.md` . Summary: **A01–A18: 17 PASS, 1 PASS WITH LIMITATION, 0 FAIL. E01–E12: 10 PASS, 2 PASS WITH LIMITATION, 0 FAIL.** (Base P0/P1 test counts referenced in that document predate this session's additions; this session's own re-run figures — 173 backend / 18 frontend tests, 2/2 Playwright e2e suites — are the current, authoritative counts and supersede any older number in that file.)

Appendix B — Model Registry Summary

Artifact	Version	Path	SHA-256
Leak classifier	1.0	models/artifacts/leak-classifier-xgb.json	0abcb0aa8992012b3e245f85c2ad4ec179bc0009b3d7f43faf267f7c544c39f9
PPE detector (active)	1.1	models/artifacts/ppe-yolo11n.pt	a6b5aedc326b2ad9118d3f5ce1f97769c746b9df92b073df0c6d62b7bacb38ae
PPE detector v1.2 candidate (rejected, not registered)	—	models/artifacts/ppe-yolo11n-v1.2-epoch7-candidate.pt (gitignored, local-only)	58365b8f1653dc0f7b8400052e6db233e84902e701e3976b86346a4e7783cbfc
GRU forecast	1.0	models/artifacts/forecast-gru.pt	4c1c5c04c541f5faa459668d1ea4567c5d3189a7eb277189e851a0a9fc8f1e02

Full record including training configuration and calibration parameters: `models/registry.json` .

Appendix C — Dataset Manifest Summary

Dataset	License	Role	Images
Ultralytics Construction-PPE	AGPL-3.0	v1.0/v1.1 training + evaluation	1132/143/141 train/val/test
Industrial Safety (Roboflow Universe)	MIT	v1.2 candidate training only (rejected)	28,119 declared; 8,000-image class-balanced training subset used

Full manifest: `models/evaluation/vision_v1.2_dataset_manifest.json` .

Appendix D — Video Source Manifest

Asset	Duration	Resolution	SHA-256	Source
demo-assets/interview_sources/clip1_helmet_alert_own_recording.mp4	13.62s	720×1280	846d3d97df759ffa121253bcc26b89b00c046578572878c0fbbad7e9a27d6f87	User's own recording
demo-assets/interview_compilation_source.mp4	17.57s	720×1280	9f7c042686bc3f01fe2717987c8ec88b254cdaa40498d4ba61c7b53501acd381	Compiled from the clip above
demo-assets/interview_compilation_annotated.mp4	17.57s	720×1280	770e4fecb3b07083ab7172c0eac1c77416393771a27df047b219ea706099029e	Real-detector-annotated compilation
deliverables/Factory_Safety_Sentinel_Interview_Demo.mp4	79.16s	1600×1000	ee7c36485cc4f12b1925bb892eed10bd775690d0a8ded2c458a26d8cbd006971	Full screen recording

Full disclosure including disqualified candidate sources: demo-assets/INTERVIEW_VIDEO_SOURCES.md .

Appendix E — Test Command Table

See §18 for the full command list. Test suite entry points: backend/tests/ (pytest), frontend/tests/ (vitest), frontend/tests/e2e/*.mjs (Playwright).

Appendix F — API Summary

Full contract: .claude/skills/factory-system-architecture/references/api-and-data-specification.md . Key routes: POST /api/v1/sensor-readings , GET /api/v1/zones/{id}/readings , GET /api/v1/zones/{id}/forecast/latest , GET /api/v1/vision/latest , GET /api/v1/vision/zones , GET /api/v1/incidents , POST /api/v1/incidents/{id}/actions , GET /api/v1/incidents/{id}/report.{json,csv} , GET /api/v1/incidents/{id}/evidence , GET/POST /api/v1/simulation/* , GET /api/v1/system/status , GET /api/v1/health/{live,ready} , WebSocket /api/v1/ws .

Appendix G — Full Decision Table

See §5 in full.

Appendix H — References and Licenses

- NIOSH CO2 pocket guide: <https://www.cdc.gov/niosh/npg/npgd0103.html> (accessed 2026-08-28)
- Ultralytics Construction-PPE dataset: <https://docs.ultralytics.com/datasets/detect/construction-ppe> (AGPL-3.0)
- Ultralytics YOLO11: <https://docs.ultralytics.com/models/yolo11> (AGPL-3.0)
- Industrial Safety dataset (Roboflow Universe): MIT license — see `models/evaluation/vision_v1.2_dataset_manifest.json`
- Natural-motion stress-test clip: Pexels video ID 5434220, "Back view of construction worker walking in safety gear on site" by Everett Bumstead — commercial use and modification permitted, no attribution required (`demo-assets/NATURAL_MOTION_SOURCE.md`)
- Interview-demonstration clip: user's own recording (`demo-assets/INTERVIEW_VIDEO_SOURCES.md`)